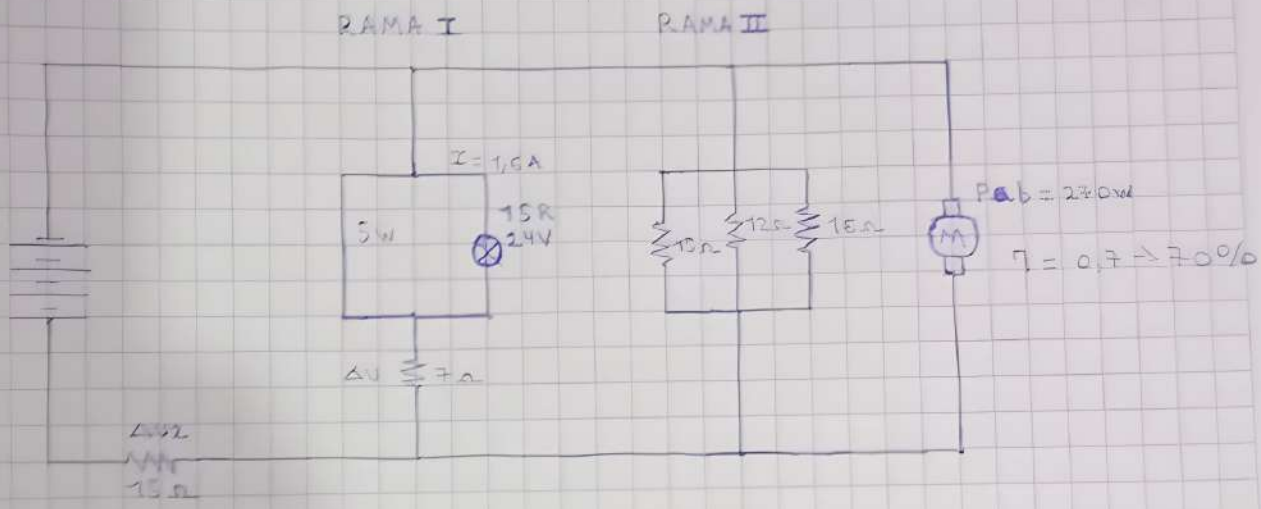
ACTIVIDAD 17

P/4 T = 2 MESES



CALCULAR

U = 39,5 V
I = 17,67 A
P = 698 W
PD = 168 W
E = 1003,72 KWH
R = 38,4 W
PP = 72 W

RAMA I

$I_{L2} = \frac{U}{R} = \frac{24V}{15\Omega} = 1,6A$

$I_{L1} = \frac{U}{P} = \frac{24V}{15W} = 0,62A$

$I_{RI} = 1,6A + 0,62A = 2,22A$

$\Delta U = IR = 2,22A \cdot 7\Omega = 15,5V$

$U_{R2} = U_L + \Delta U = 24V + 15,5V = 39,5V$

$P_L = U \cdot I = 24V \cdot 1,6A = 38,4W$

RAMA II

$I_1 = \frac{V}{R} = \frac{39,5V}{15\Omega} = 2,63A$ $I_2 = \frac{U}{R} = \frac{39,5V}{12\Omega} = 3,29A$

$I_3 = \frac{V}{R} = \frac{39,5V}{16\Omega} = 2,46A$ $I_{RII} = 2,63A + 3,29A + 2,46A = 8,38A$

RAMA III → $I = \frac{Pab}{U} = \frac{240W}{39,5V} = 6,07A$

PU =

$PU = 240W \cdot 0,7 = 168W$

$\eta = \frac{PU}{Pab} \Rightarrow PU = Pab \cdot \eta$ $P_D = 168W \cdot 1HR$